

Sociogenomics

Gene-environment interplay

Nicola Barban



What is gene-environment interplay?

- Gene-environment interplay is the phenomenon of
 - A. gene-environment interaction (GxE) and
 - B. gene-environment correlation (rGE).

Gene-environment correlation

- G-E correlation (r_{GE}) measures **whether there are different allele frequencies in different environments**
- is the process by which an individual's genotype influences or is associated with exposure to the environment or in other words, how genes and the environment operate in tandem.

Gene-environment interactions

- GxE studies whether the **effect of the genotype on the phenotype varies across different environments.**
- Gene-environment interaction (GxE) defines an interplay between a gene and an environmental factor in which **the effect of the gene on a phenotype is modifiable by the environment and vice versa.**

What is the environment?

- In the context of genetic research, **Environment (E)**, however, can adopt multiple forms.
- Environmental risk factor can be an exposure, either physical (e.g., radiation, temperature), chemical (e.g., polycyclic aromatic hydrocarbons), or biological (e.g., a virus); a behaviour pattern (e.g., late age at first pregnancy); or a "life event" (e.g., job loss, injury).
- In genetics E is **in practice everything that is non-genetic.**

History of GxE

- A. Twins studies and heritability
- B. Candidate gene studies
- C. Polygenic scores

Twin studies and GxE

- The ACE model assumes no GXE effects
- However, we can estimate the model in **different environments**
- Typically different birth cohorts or different family background
- Sometimes this can be estimated within SEM models

Turkheimer, E., Haley, A., Waldron, M., D'Onofrio, B., & Gottesman, I.I. (2003). Socioeconomic status modifies heritability of IQ in young children. *Psychological Science*, 14(6), 623-628.

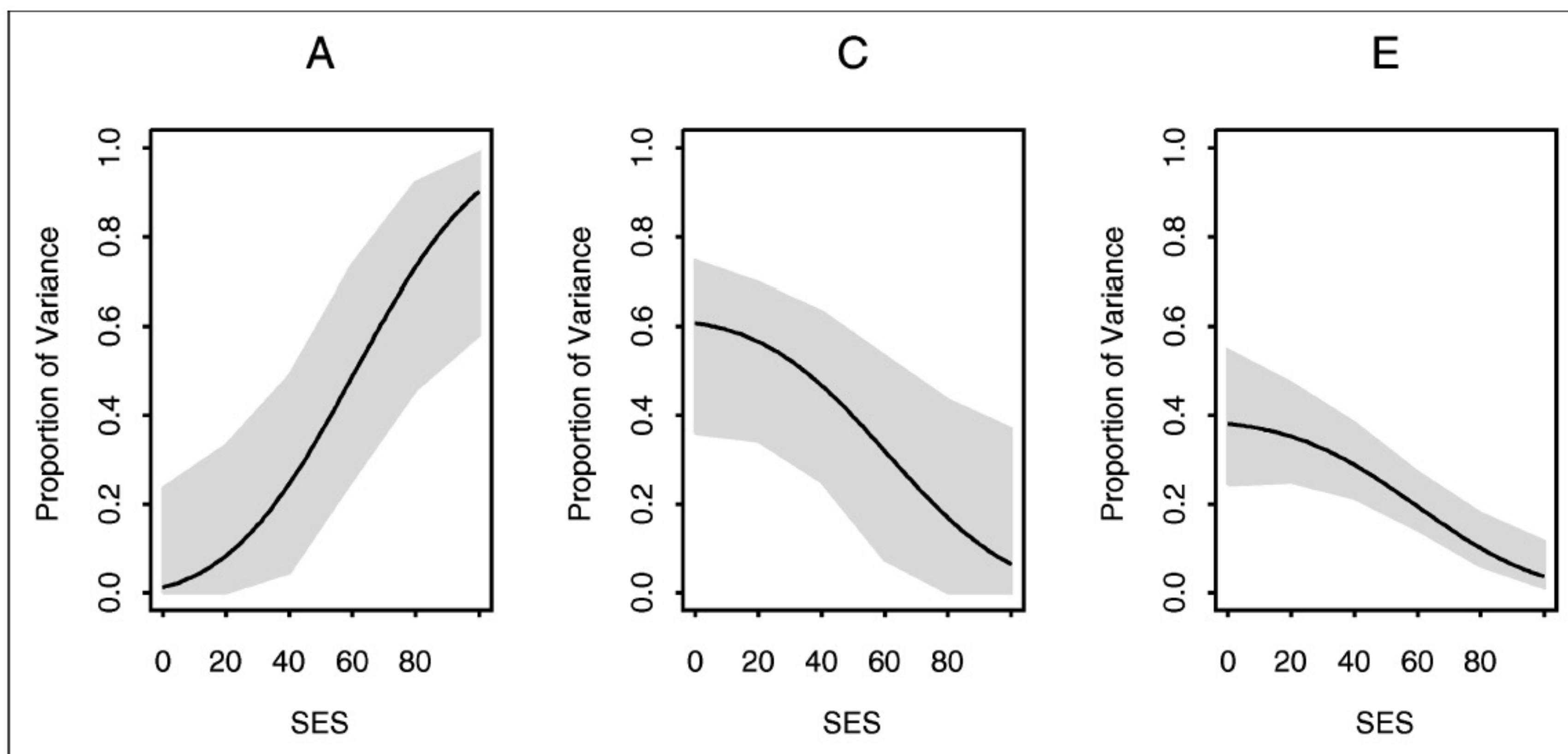
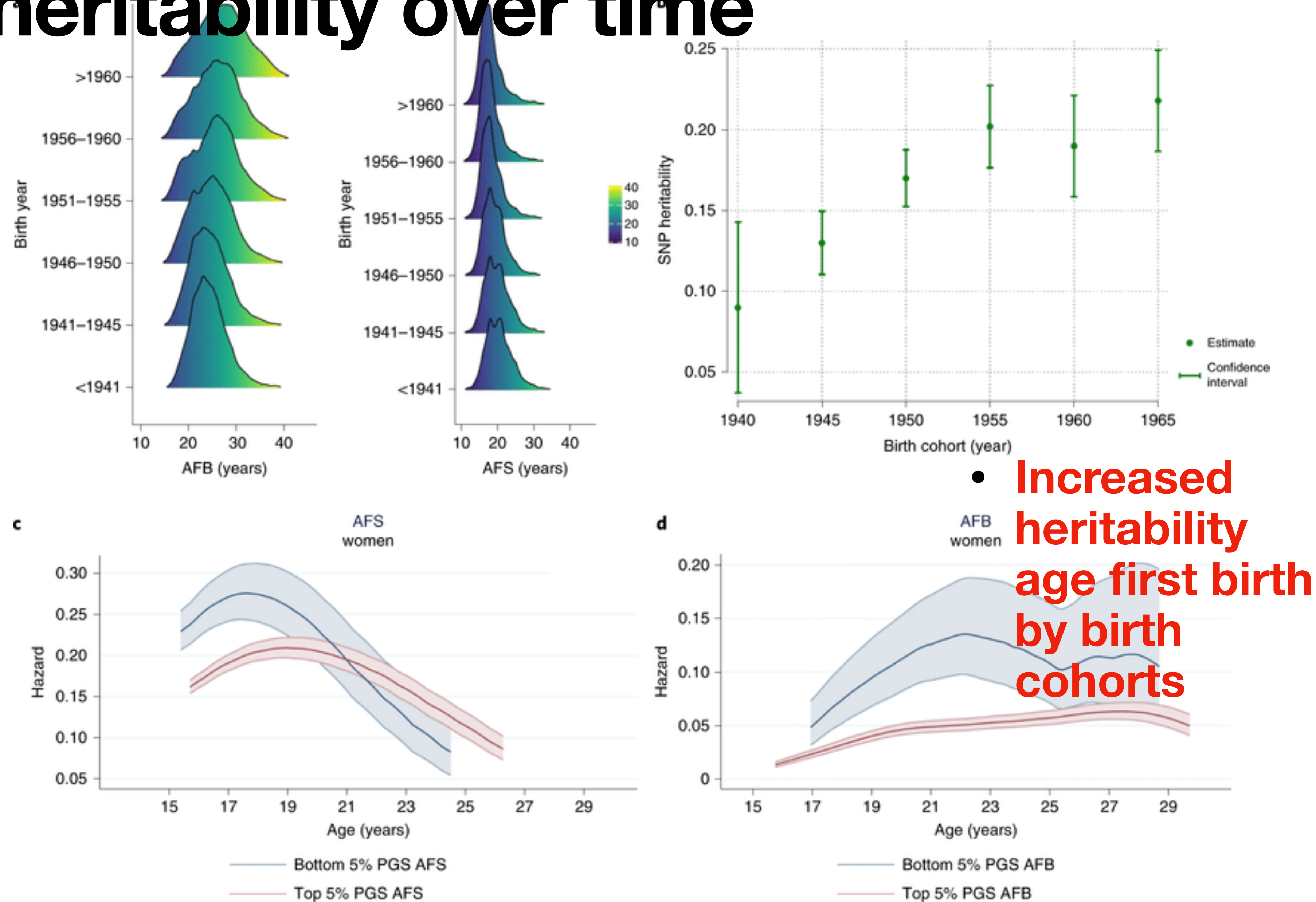


Fig. 3. Proportion of total Full-Scale IQ variance accounted for by A, C, and E plotted as a function of observed socioeconomic status (SES). Shading indicates 95% confidence intervals.

Scarr- Rowe Hypothesis

- The Scarr–Rowe hypothesis, refers to the proposed **moderating effect of low socioeconomic status on the heritability of children's IQ**. According to this hypothesis, lower socioeconomic status and greater exposure to social disadvantage during childhood leads to a decrease in the heritability of IQ, as compared to children raised in more advantaged environments
- A 2015 meta-analysis found that the effect was predominant in the United States while less evident in societies with robust child welfare systems

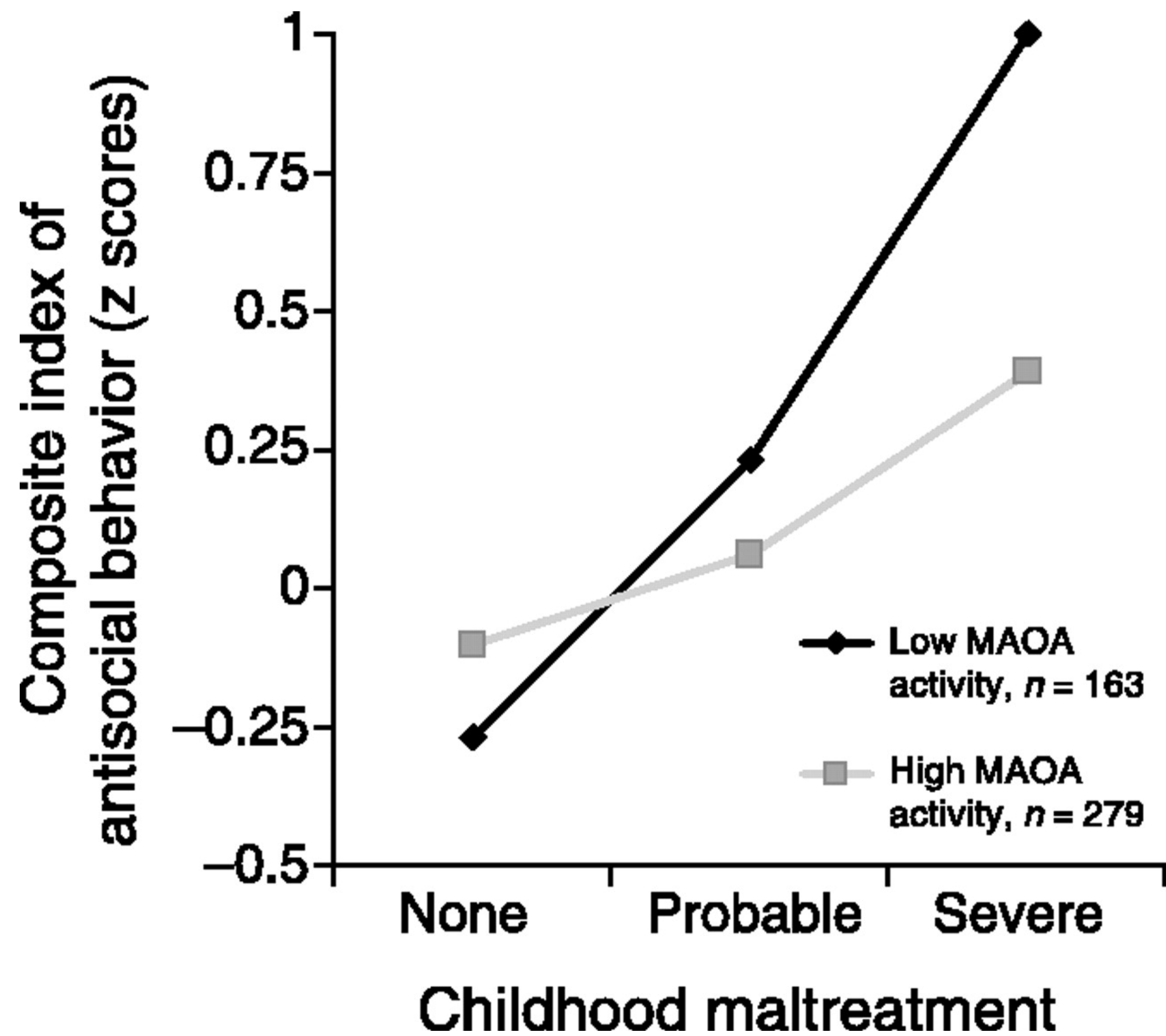
Change of heritability over time



Candidate gene studies and GxE

- **candidate gene studies** in the 2000s, focussed on pre-defined loci of interest based on what was thought to be a priori knowledge of the loci's biological function or impact on the trait being examined.

“Role of Genotype in the Cycle of
Violence in Maltreated Children”, Caspi
et al. 2002 Science



FTO and Obesity



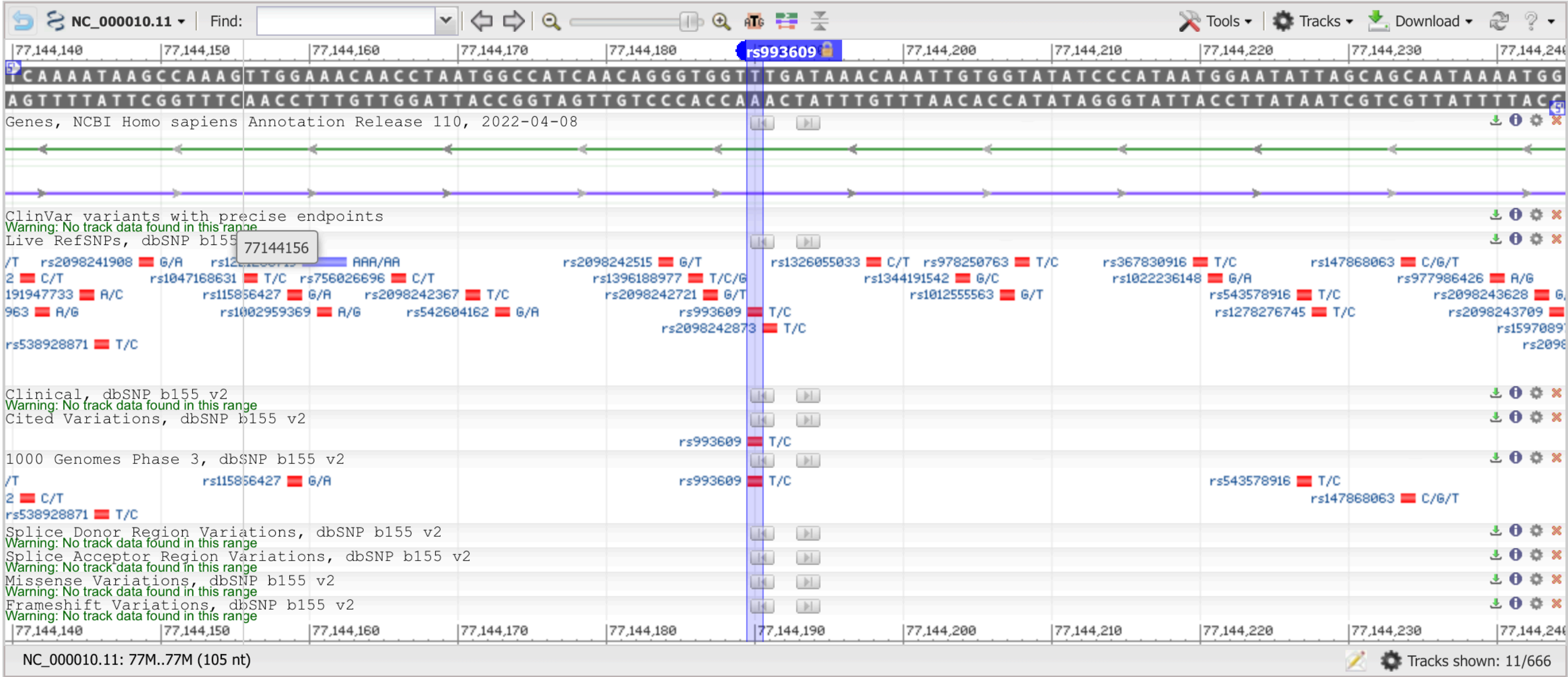
Cohort of birth modifies the association between FTO genotype and BMI

James Niels Rosenquist^{a,1}, Steven F. Lehrer^{b,c}, A. James O'Malley^d, Alan M. Zaslavsky^e, Jordan W. Smoller^f, and Nicholas A. Christakis^{g,h,i,j}

^aDepartment of Psychiatry, Massachusetts General Hospital, Boston, MA 02114; ^bSchool of Policy Studies and Department of Economics, Queens University, Kingston, Ontario, Canada K7L 3N6; ^cNational Bureau of Economic Research USA, Cambridge, MA 02138; ^dThe Dartmouth Institute for Health Policy and Clinical Practice, Geisel School of Medicine, Dartmouth College, Hanover, NH 03755; ^eDepartment of Health Care Policy, Harvard Medical School, Boston, MA 02115; ^fPsychiatric and Neurodevelopmental Genetics Unit, Center for Human Genetic Research, Massachusetts General Hospital, Boston, MA 02114; ^gDepartment of Sociology, Yale University, New Haven, CT 06520; ^hDepartment of Medicine, Yale University, New Haven, CT 06520; ⁱDepartment of Ecology and Evolutionary Biology, Yale University, New Haven, CT 06520; and ^jYale Institute for Network Science, Yale University, New Haven, CT 06520

Edited by Kenneth W. Wachter, University of California, Berkeley, CA, and approved November 11, 2014 (received for review June 25, 2014)

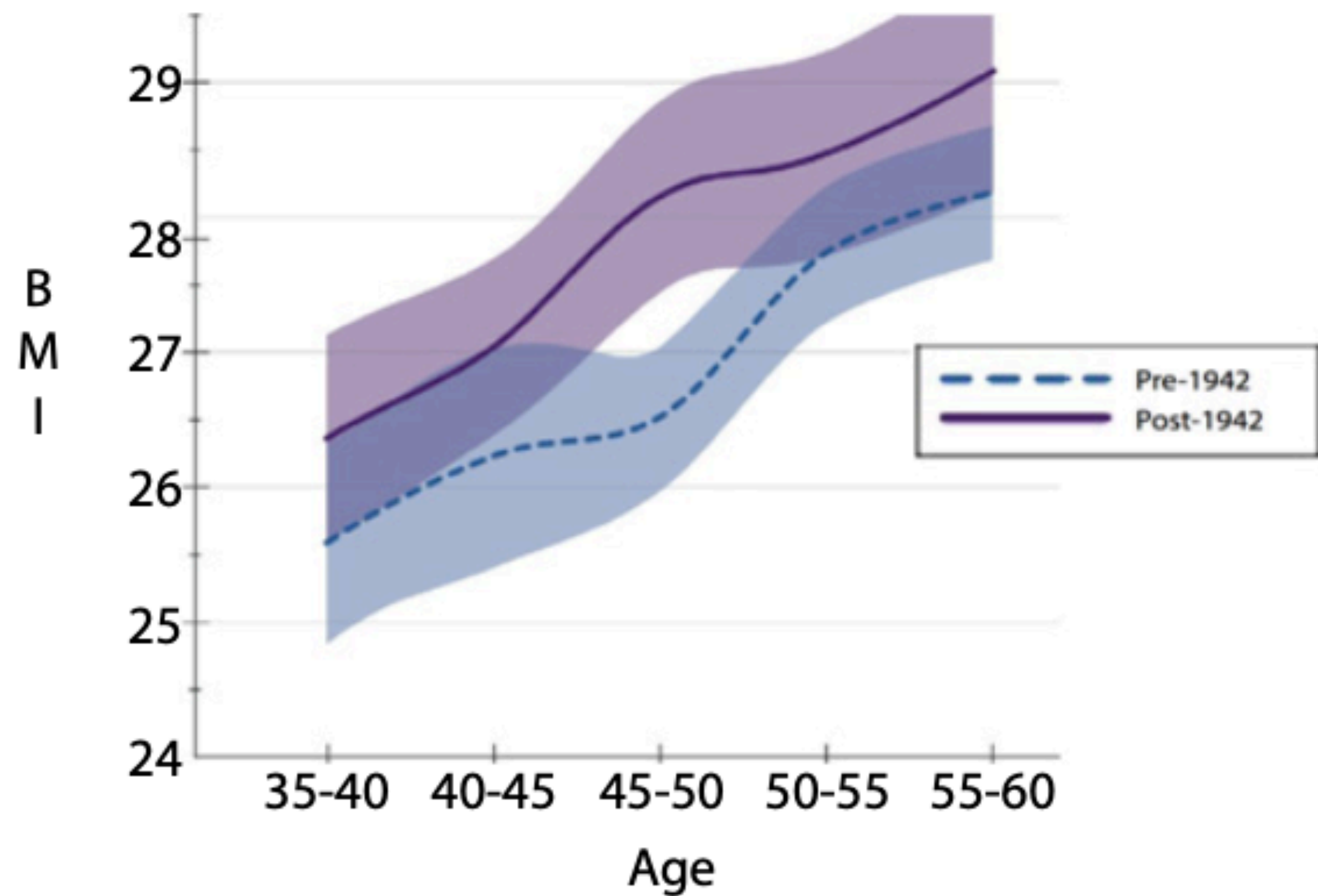
- “we use longitudinal data from the Framingham Heart Study collected over 30 y from a geographically relatively localized sample to test whether the well-documented association between the rs993609 variant of the FTO (fat mass and obesity associated) gene and body mass index (BMI) varies across birth cohorts, time period, and the lifecycle. “



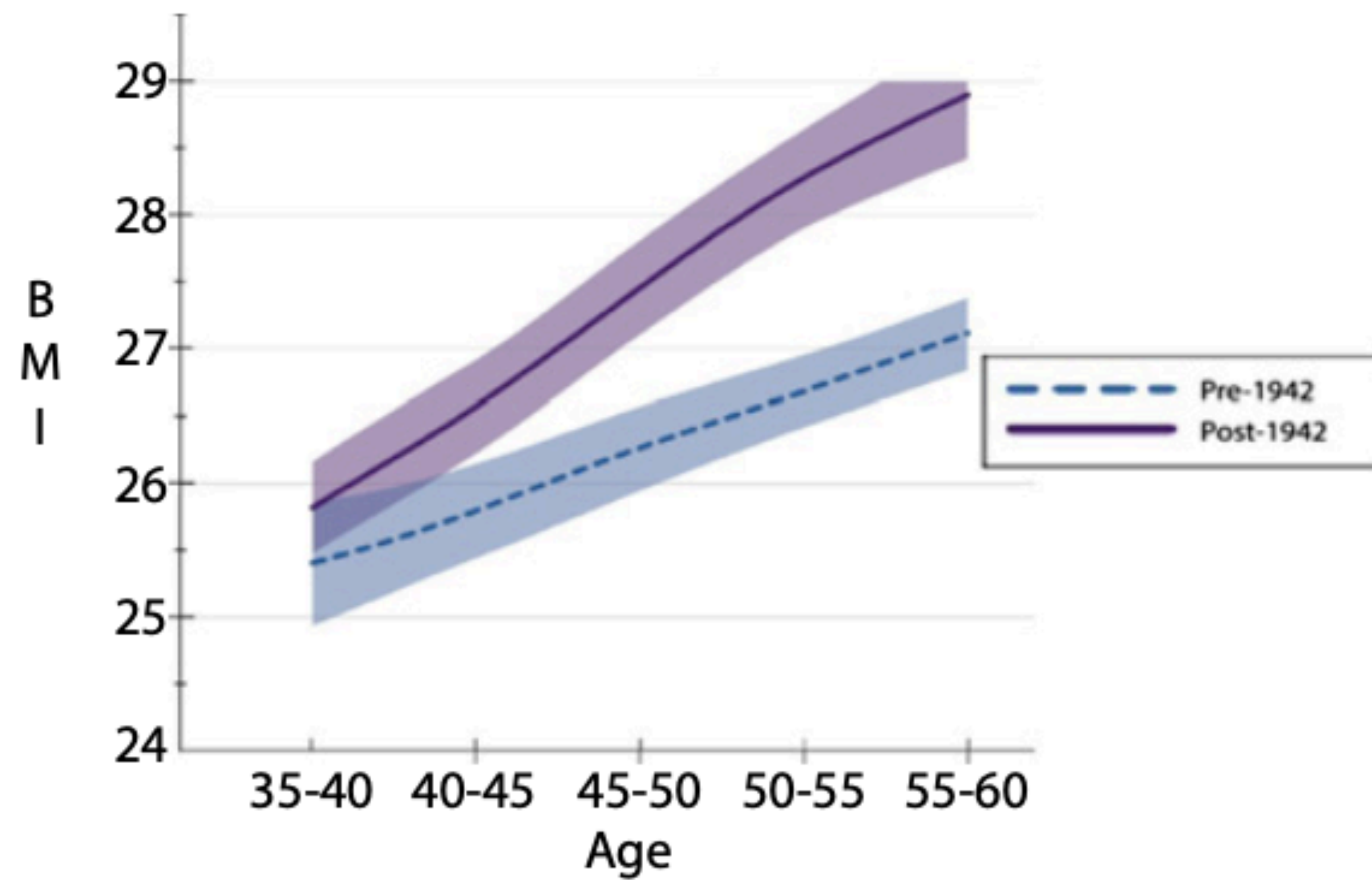
Search:

Population	Group	Sample Size	Ref Allele	Alt Allele
Total	Global	18890	T=0.53515	C=0.46485
European	Sub	14286	T=0.51323	C=0.48677
African	Sub	2946	T=0.5642	C=0.4358
African Others	Sub	114	T=0.535	C=0.465
African American	Sub	2832	T=0.5653	C=0.4347
Asian	Sub	112	T=0.955	C=0.045
East Asian	Sub	86	T=0.95	C=0.05
Other Asian	Sub	26	T=0.96	C=0.04

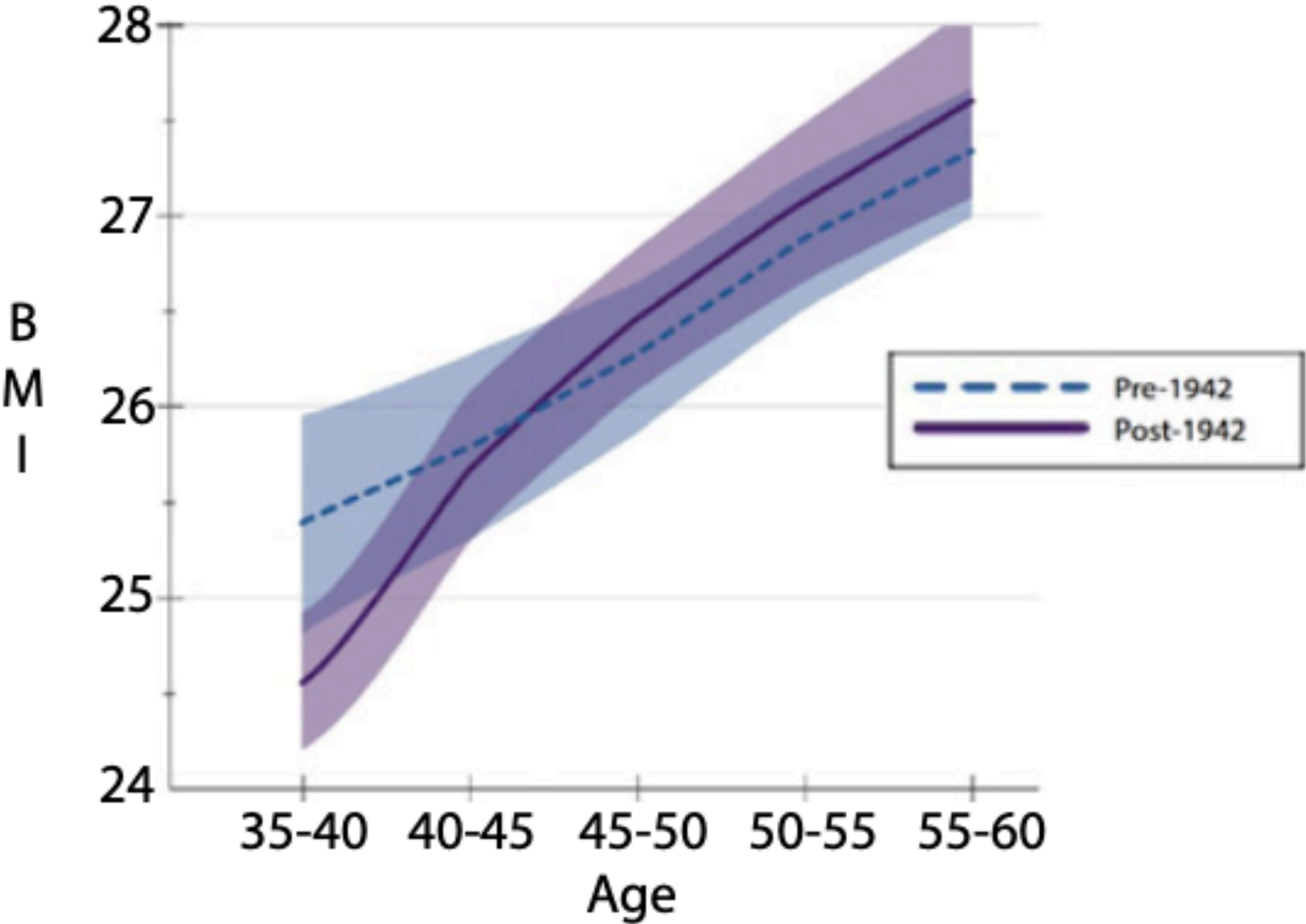
AA-FTO genotype



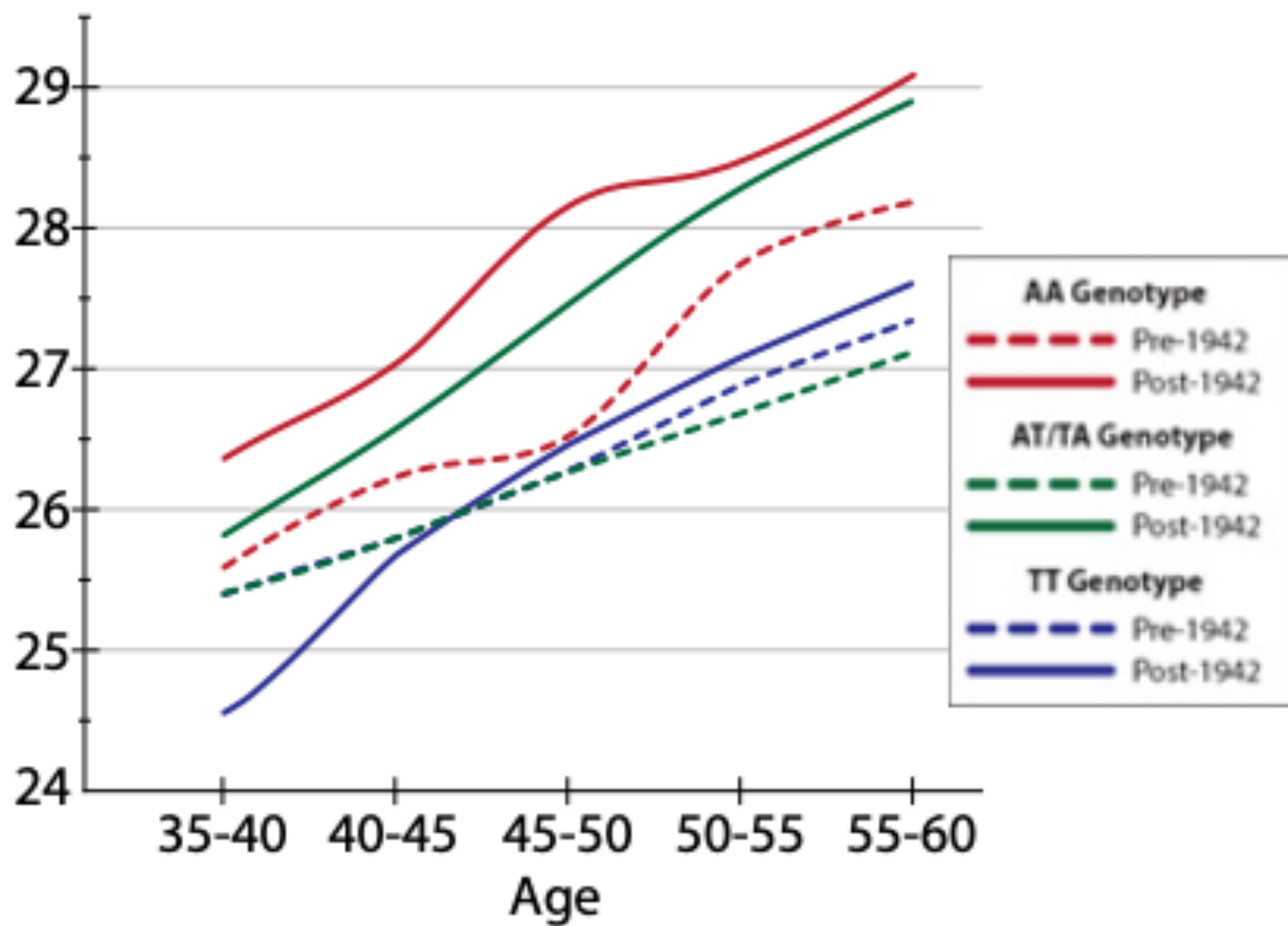
AT/TT-FTO genotype



TT-FTO genotype



B
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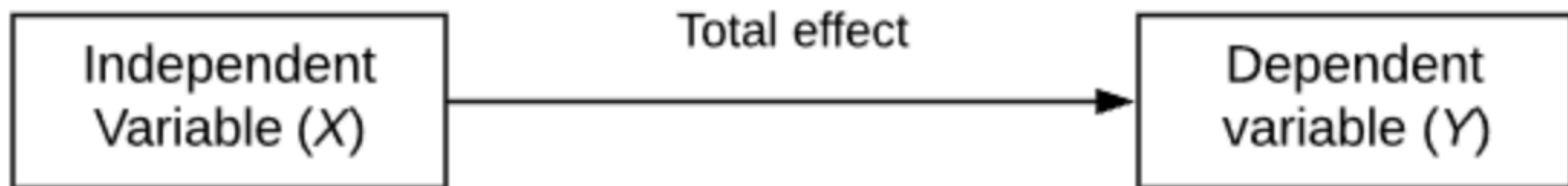


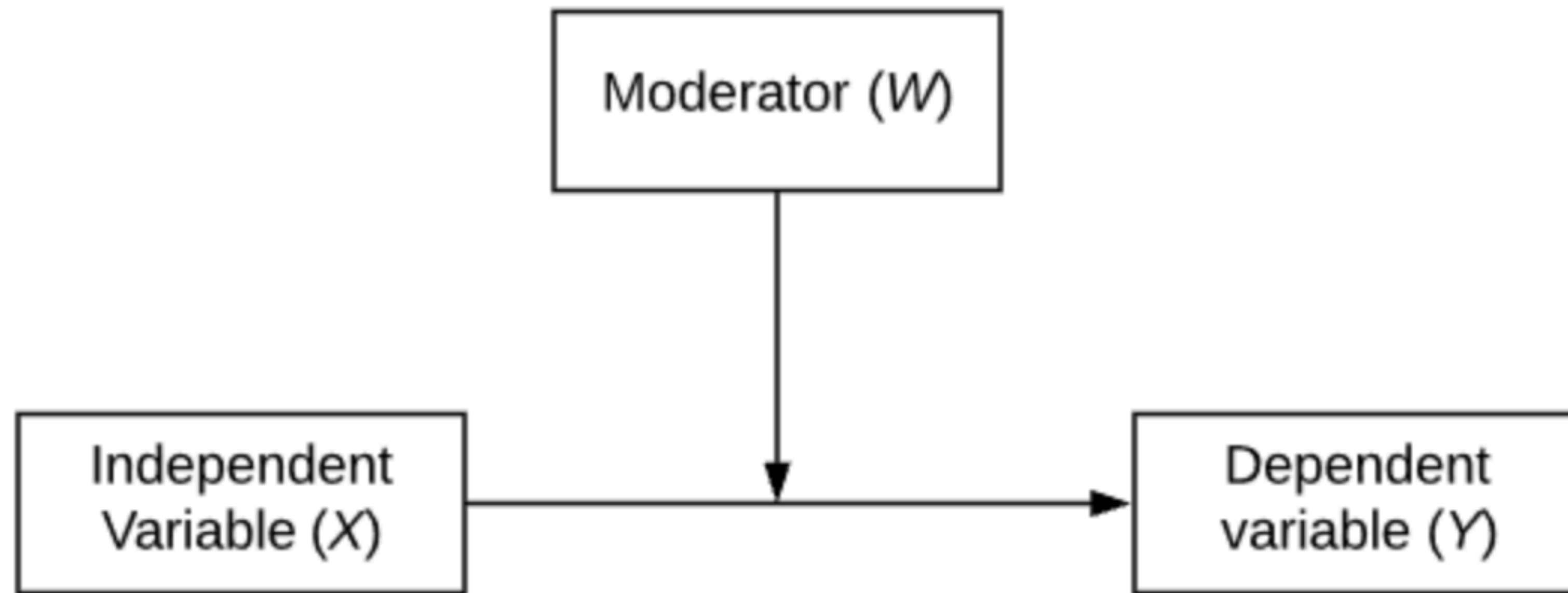
Interpretation

- “Our results suggest that the well-documented rise in BMI in the United States over the past 40 y may have been disproportionately **driven by individuals for whom genetic factors interacted with environmental changes encountered in their development due to their era of birth**—in this case, being born later.
- Environmental influences on the rise in obesity did indeed occur after the early 1940s, including technological advances **reducing energy expenditure at work** as well as **increases in the caloric content of processed foods**

Genes as moderators and mediator

- A mediating variable (or **mediator**) explains the process through which two variables are related, while a moderating variable (or **moderator**) affects the strength and direction of that relationship.





Why GxE effects?

dandelions can thrive in most environments.

“**orchids**”, who are more exquisite and unusual and have a higher potential than dandelions – but for this to be realised they require a particular environment and careful gardening.

The Orchid and the Dandelion by W Thomas Boyce review – which are you?

Are you sensitive or resilient? This study by a paediatric health expert considers why children with the greatest potential are also the ones most likely to falter



Orchids or dandelions? ... four fifths of children can thrive in most environments. Photograph:



GxE in post GWAS era

- **It is common to use polygenic scores to study GxE**
- individual-level genetic endowments (that are fixed at birth)
- When we use more powerful genetic predictors, we generally have less of a sense for the specific biological mechanisms

Genetic risk and age

Obesity

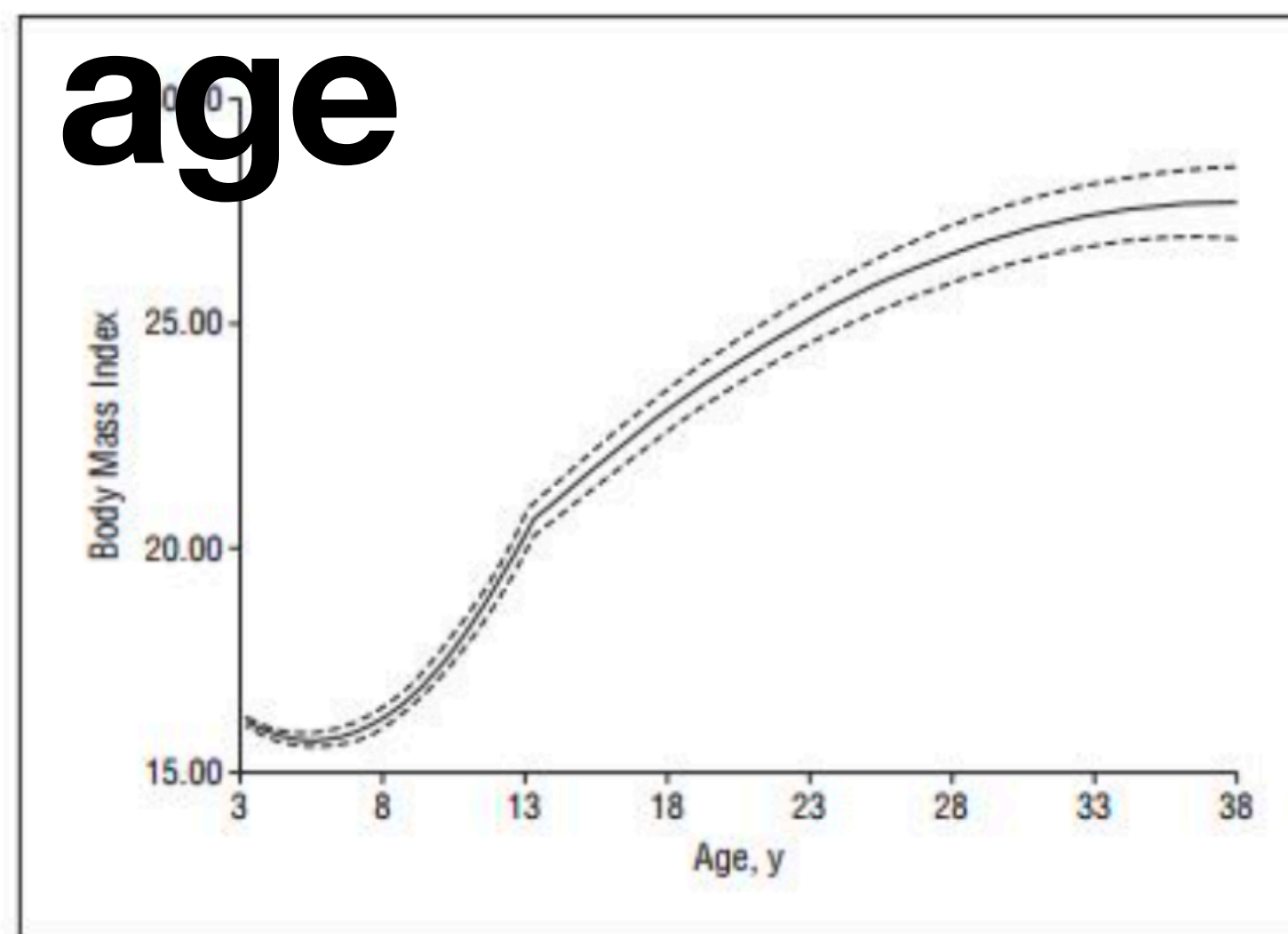


Figure 2. Life-course growth curves for children with high, low, and average genetic risk scores (GRSs). Individuals with higher-obesity GRSs were larger and grew more rapidly as children and adults. The solid line represents the population mean trajectory (average genetic risk). Dashed lines are for subgroups within 1 SD of the GRS (high and low genetic risk). Trajectories were derived from the life-course growth model (intercept fitted at 13 years of age; linear and quadratic slopes fitted during ages 3-13 years and 13-38 years), including intercept and linear slope effects for the GRS. Analyses included 856 individuals of European descent. Body mass index is calculated as weight in kilograms divided by height in meters squared.

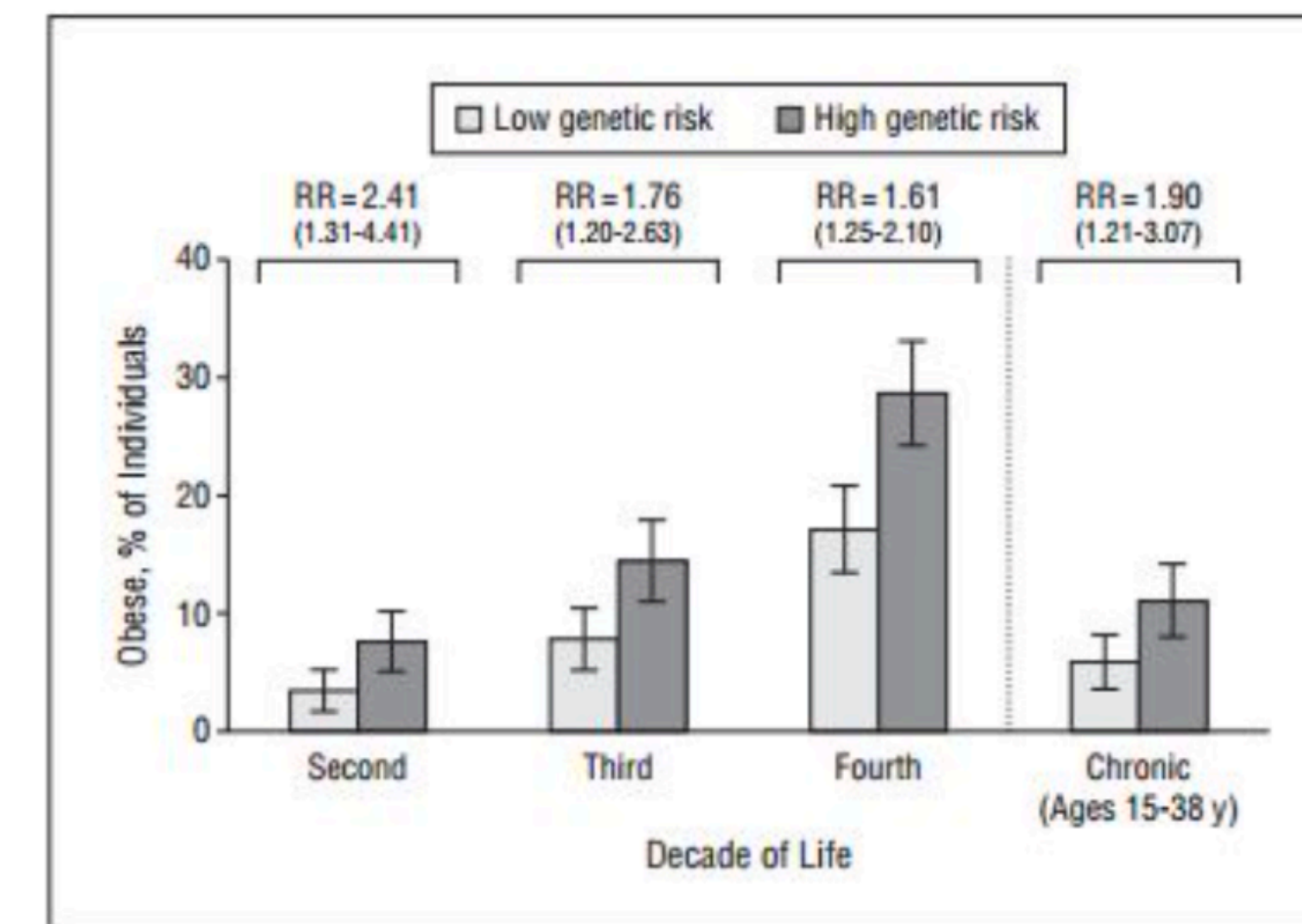


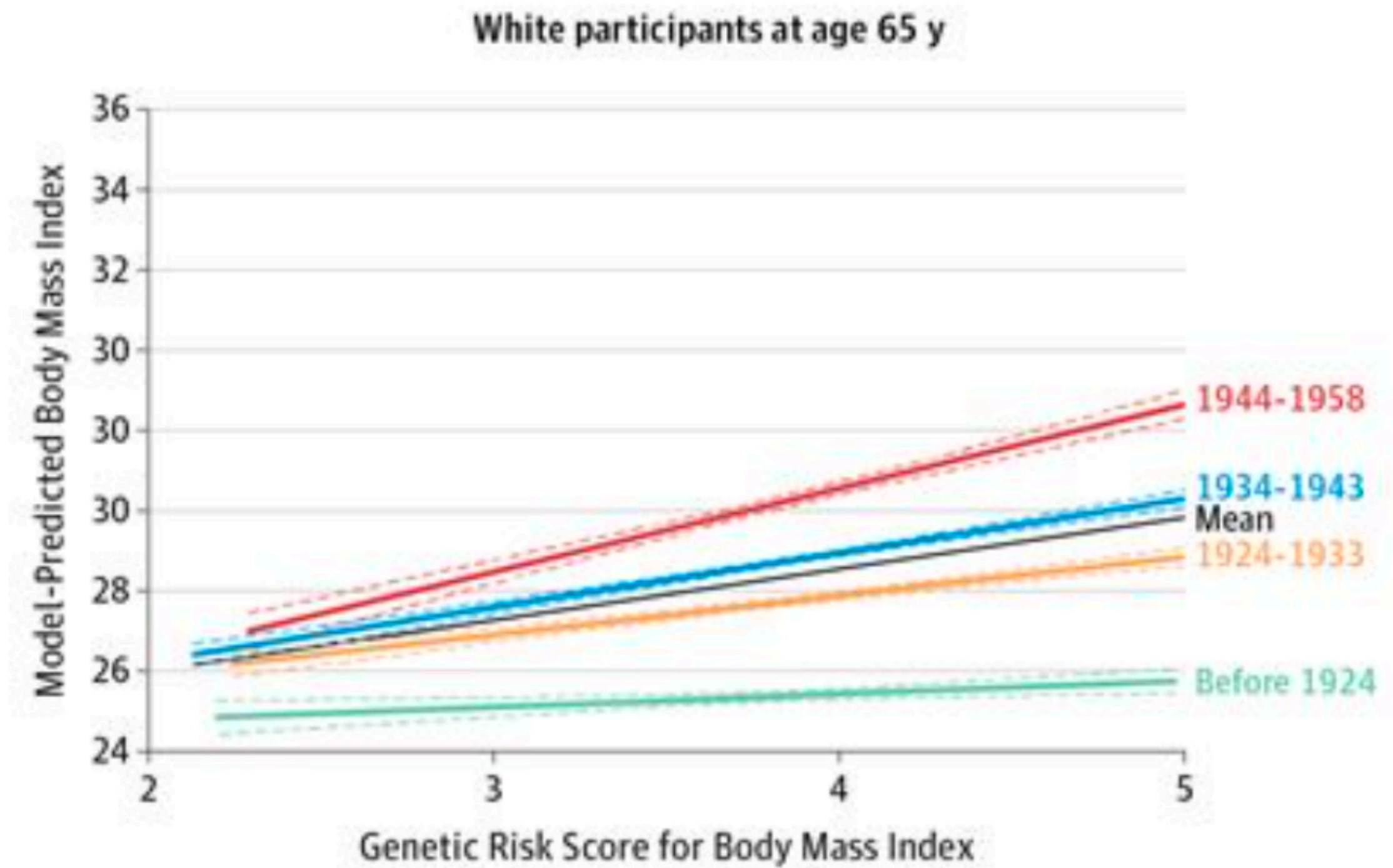
Figure 3. Obesity prevalence among low and high genetic risk cohort members in their second, third, and fourth decades of life and chronically across ages 15 to 38 years. Individuals with higher genetic risk scores (GRSs) were more likely to be obese across 2 decades of adult follow-up. Error bars and numbers in parentheses reflect 95% CIs. The GRS was dichotomized at the sample mean to create low and high genetic risk categories. Relative risks (RRs) (95% CIs) are reported from Poisson regression models adjusted for sex that included the 856 individuals of European descent in the analysis sample.

Polygenic Risk, Rapid Childhood Growth, and the Development of Obesity

Evidence From a 4-Decade Longitudinal Study

Daniel W. Belsky, PhD; Terrie E. Moffitt, PhD; Renate Houts, PhD; Gary G. Bennett, PhD; Andrea K. Biddle, PhD; James A. Blumenthal, PhD; James P. Evans, MD, PhD; HonaLee Harrington, BA; Karen Sugden, PhD; Benjamin Williams, BS; Richie Poulton, PhD; Avshalom Caspi, PhD

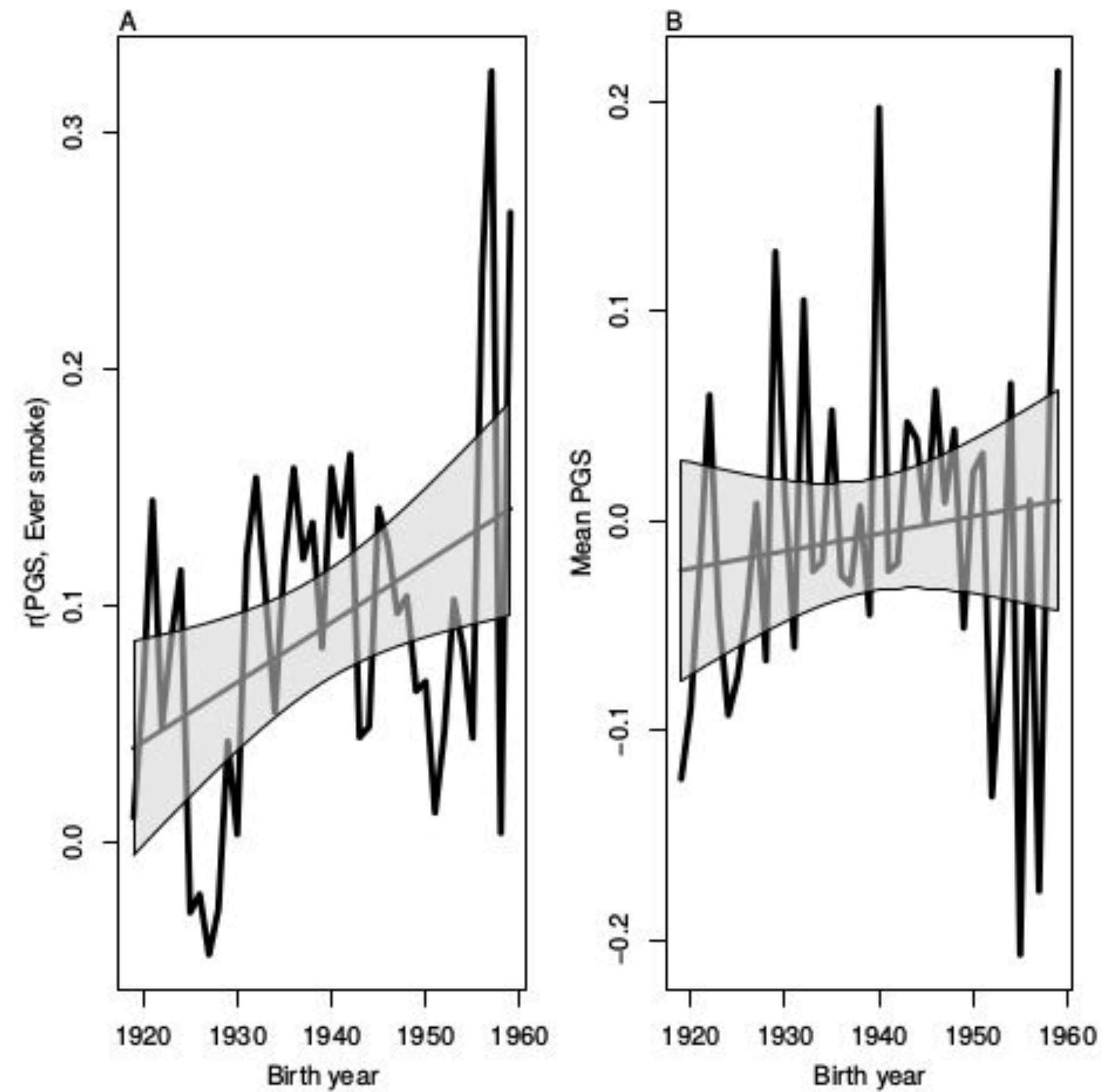
BMI



Association of a Genetic Risk Score With Body Mass Index Across Different Birth Cohorts

Stefan Walter, PhD¹; Iván Mejía-Guevara, PhD²; Karol Estrada, PhD³; [et al](#)

Smoking



Behav Genet
DOI 10.1007/s10519-015-9731-9



ORIGINAL RESEARCH

Cohort Effects in the Genetic Influence on Smoking

Benjamin W. Domingue¹ · Dalton Conley² · Jason Fletcher³ · Jason D. Boardman⁴

The model

- To estimate GxE needs some assumptions

$$Y_i = \alpha + \beta G_i + \gamma E_i + \rho(G_i \times E_i) + \epsilon_i$$

The model

- To estimate GxE needs some assumptions

$$Y_i = \alpha + \beta G_i + \gamma E_i + \rho(G_i \times E_i) + \epsilon_i$$

$$G, E \perp \epsilon$$

**G and E need to be independent
of the error term, aka exogenous**

Exogenous environment

- People choose their environments, and unobserved variables can affect the estimate of E and GxE
- Ideally you would assign randomly environment to subjects
- This is in most case not possible
- Possible solutions are to use the so-called “natural experiments”

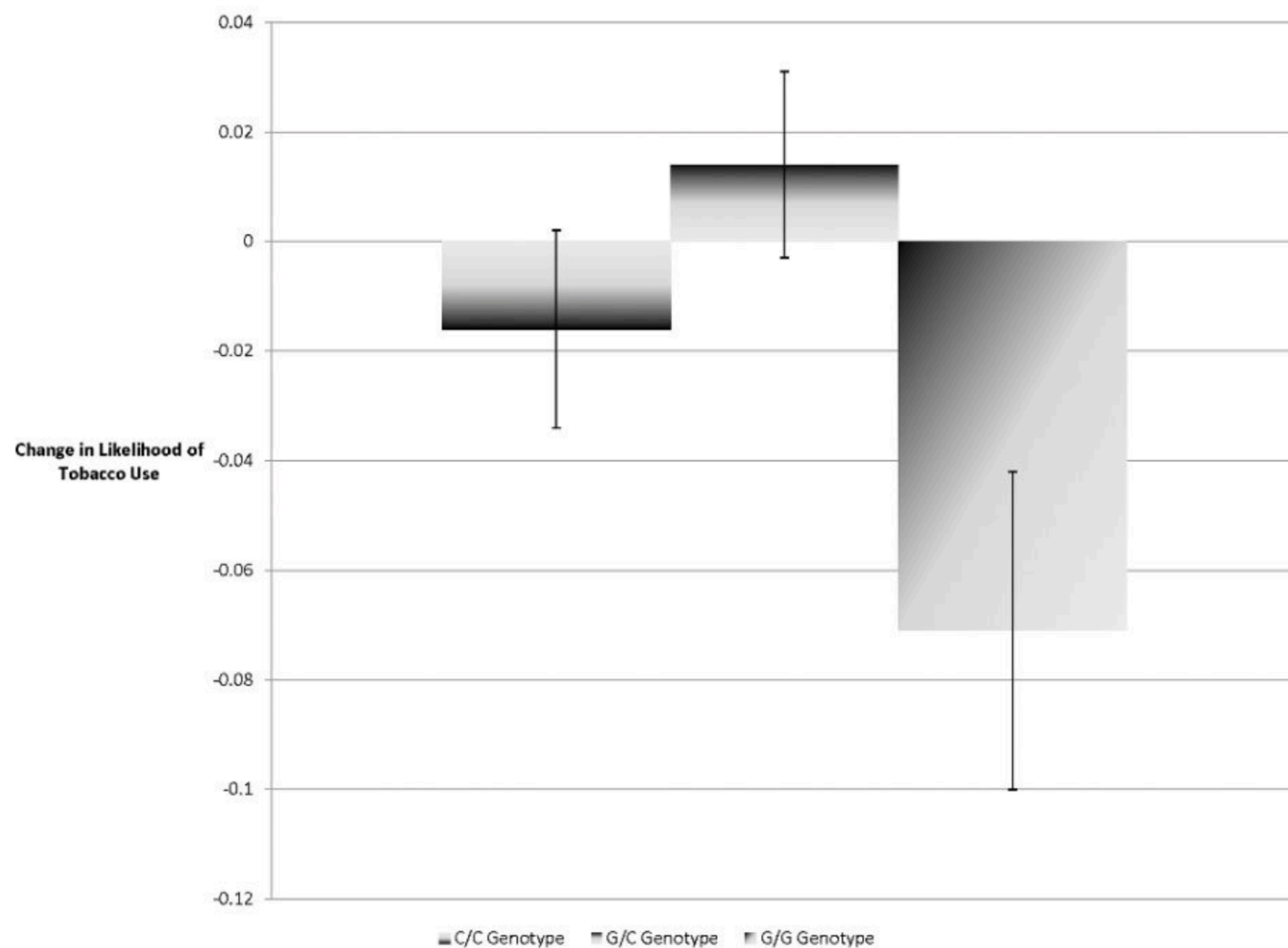
Smoking & Taxes

Why Have Tobacco Control Policies Stalled? Using Genetic Moderation to Examine Policy Impacts

Jason M. Fletcher 

Published: December 5, 2012 • <https://doi.org/10.1371/journal.pone.0050576>

Change in cigarette taxes and genetics



Notes: Author's calculation from NHANES Data. **Results from three separate regression analyses estimating the association between state level tobacco tax rates and tobacco use based on *CHRNA6* genotype (C/C, G/C, G/G).** Among the 1,278 individuals with C/C genotype, the estimated tax effect was not significant ($b=-0.016$, $SE=0.018$, $z=.89$, $P=0.37$). Among the 2,328 individuals with the G/C genotype, the estimated tax effect was not significant ($b=0.014$, $SE=0.017$, $z=.82$, $P=.41$). However, among the 2,572 individuals with the G/G genotype, the estimated tax effect was statistically significant ($b=-0.071$, $SE=0.029$, $z=2.44$, $P=0.0148$). These effects were statistically different based on genotype (see full results in [Table 2](#)). **This interaction showed that only adults with the G/G genotype respond to tobacco taxes in the manner predicted by economic theory.** Robust standard error bars clustered at the state level. Sample weights were used in order to produce nationally representative estimates.

Price elasticity depends on genotype

Genetic Heterogeneity in Depressive Symptoms Following the Death of a Spouse: Polygenic Score Analysis of the U.S. Health and Retirement Study

Benjamin W. Domingue, Ph.D., Hexuan Liu, Ph.D., Aysu Okbay, Ph.D., Daniel W. Belsky, Ph.D.

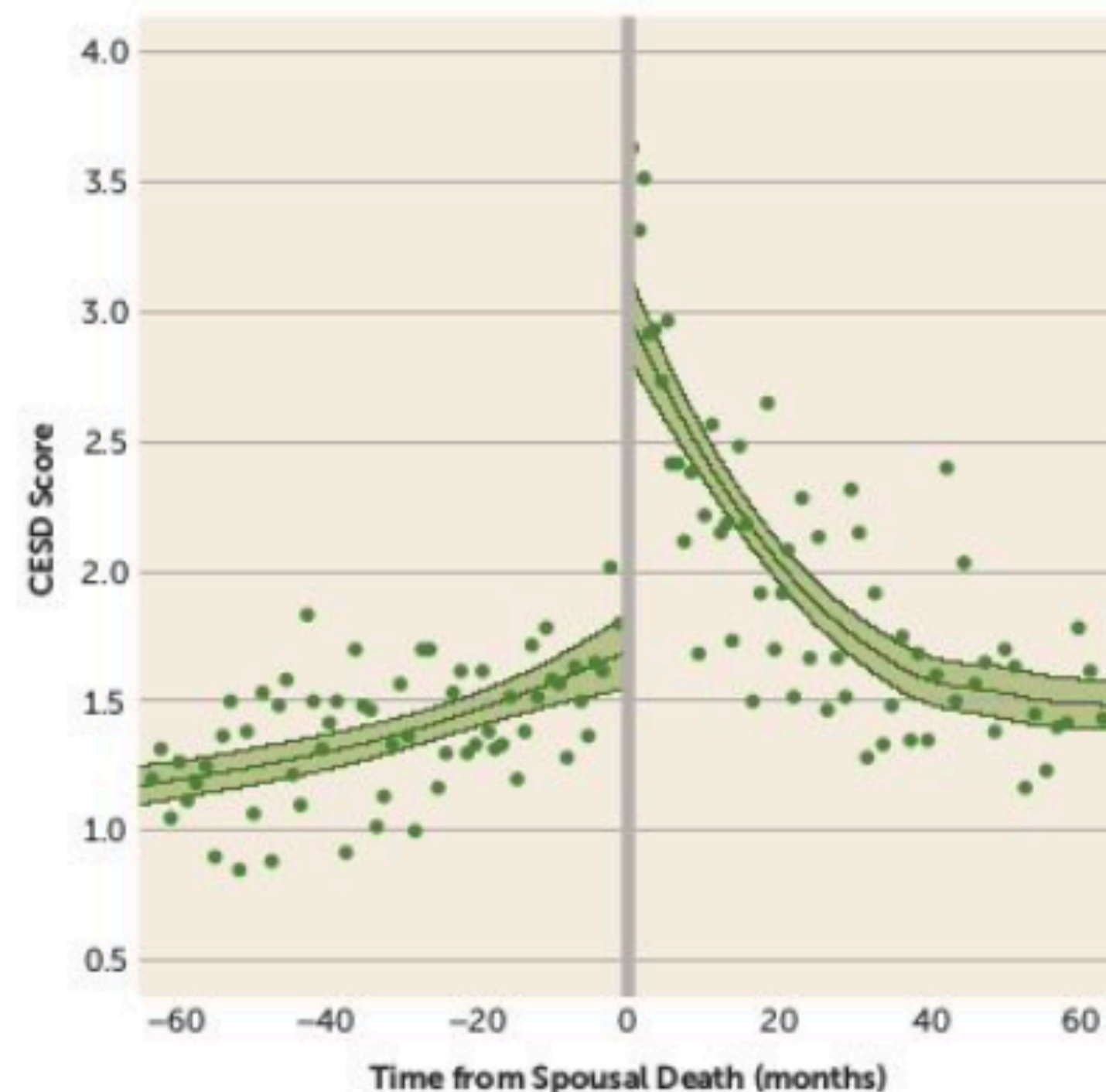
Received: November 02, 2016

Accepted: February 09, 2017

Published online: March 24, 2017 | <http://dx.doi.org/10.1176/appi.ajp.2017.16111209>

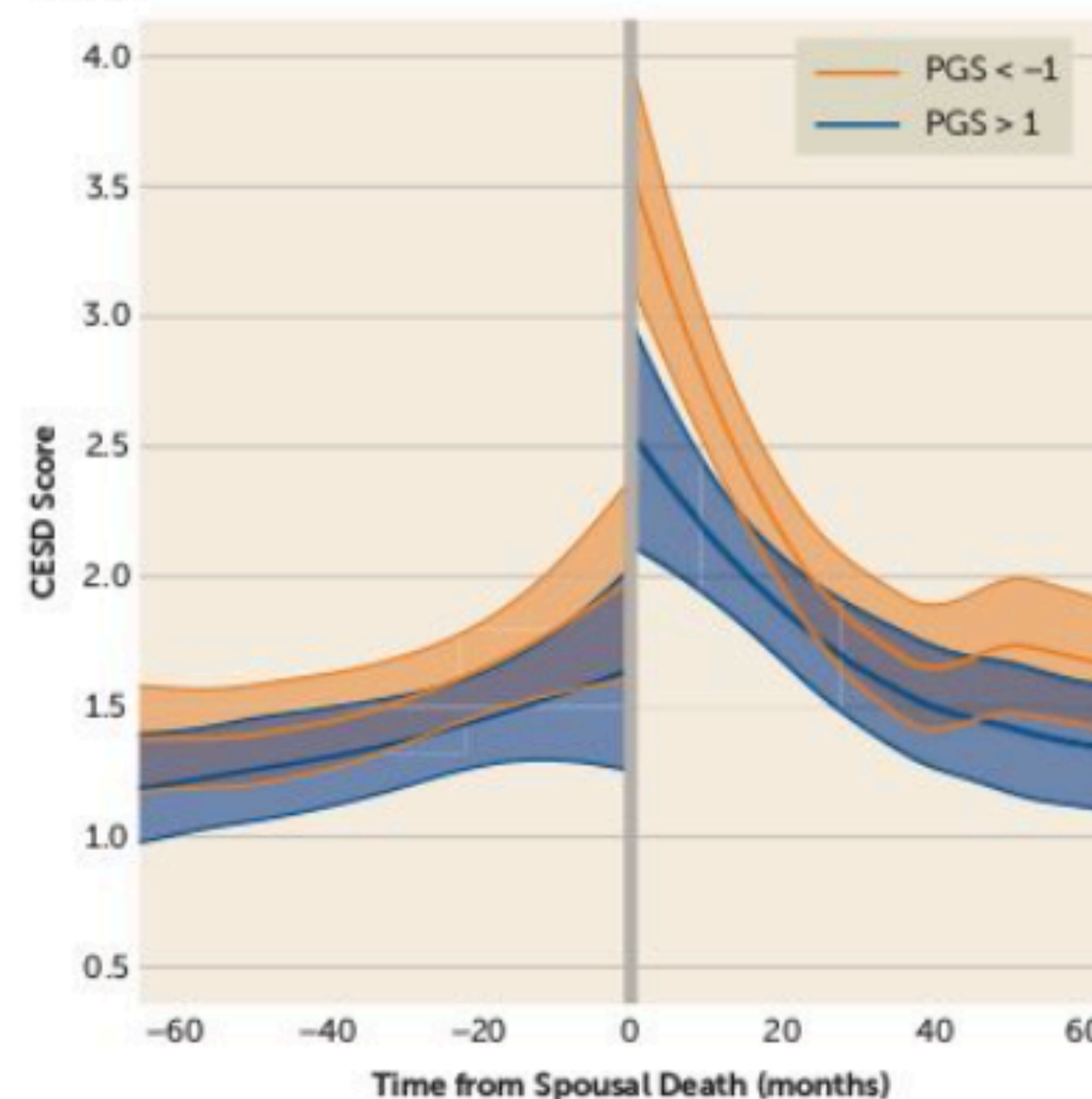
Depression after the death of a spouse

FIGURE 2. Trajectories of Depressive Symptoms in the Months Surrounding the Death of a Spouse^a



^a The scatterplot and local regression plot show depressive symptoms (Center for Epidemiological Studies Depression [CESD] scores) by month of measurement relative to the death of a spouse. The plot reflects data from Health and Retirement Study respondents who experienced spousal death (N=1,647; 14,309 total observations). Plotted points show mean x- and y- coordinates for bins of about 60 observations.

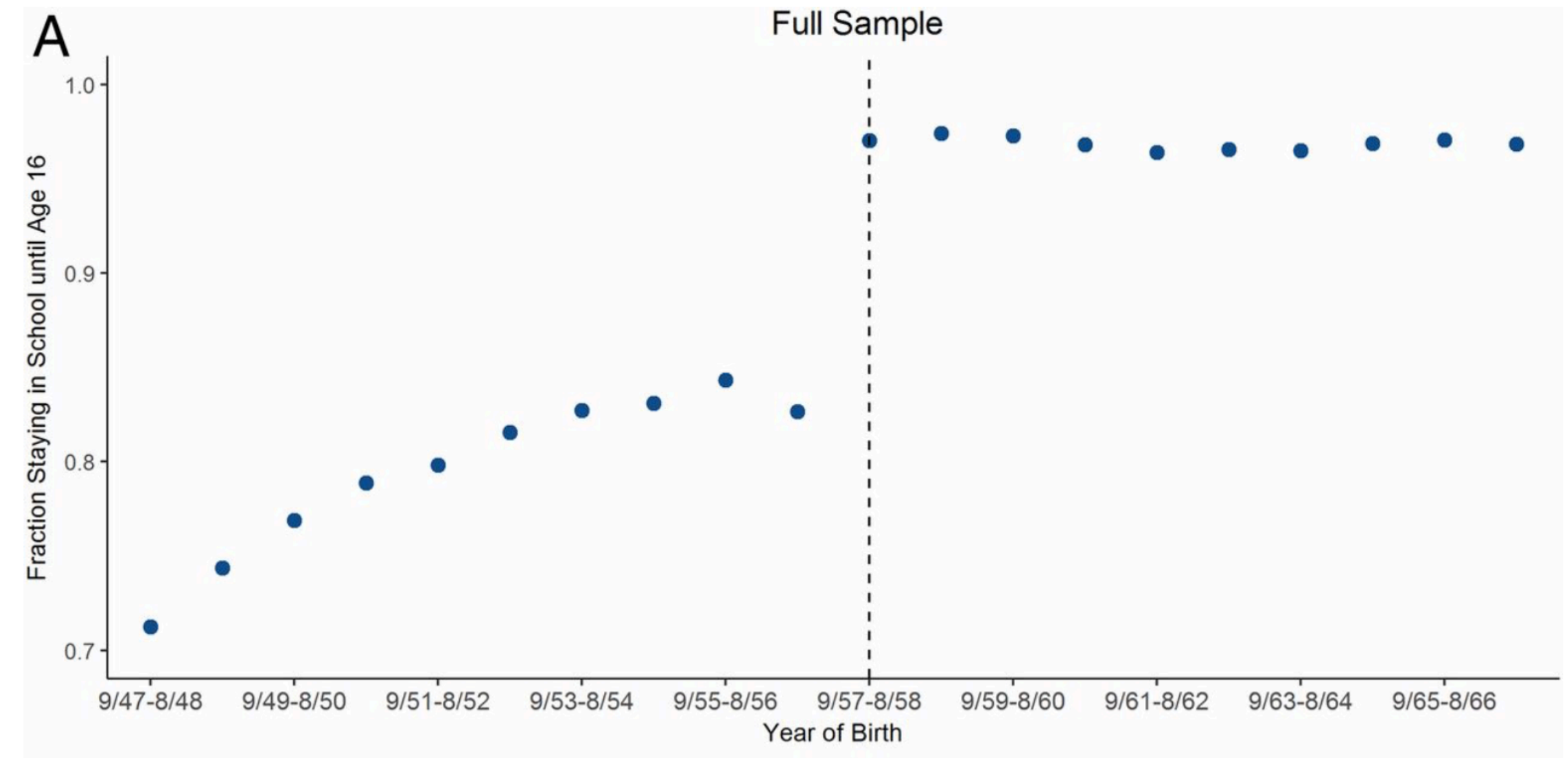
of Depressive Symptoms for Health and Retirement Study (HRS) Respondents With Low Subjective Well-Being Polygenic Scores and High Subjective Well-Being Polygenic Scores^a



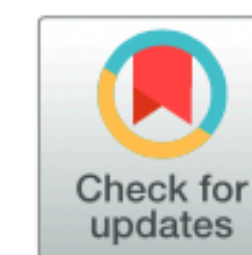
^a The graph shows local regression plots of depressive symptoms (Center for Epidemiological Studies Depression [CESD] scores) by month of measurement relative to the death of a spouse. Trajectories for respondents with low subjective well-being polygenic scores are graphed in red (1 or more standard deviations below the mean, N=255 individuals with 2,164 observations). Trajectories for HRS respondents with high subjective well-being polygenic scores are graphed in blue (1 or more standard deviations above the mean, N=242 individuals with 2,048 observations). PGS=polygenic score.

Raise of school leaving age ROSLA

- In March 1959, UK raised the the school leaving age to 16
- Individuals born after September 1, 1957 had to stay in school until age 16
- It is then possible to compare individuals around the introduction of the reform
- This is called a **regression discontinuity design**



Can education reduce health inequalities?



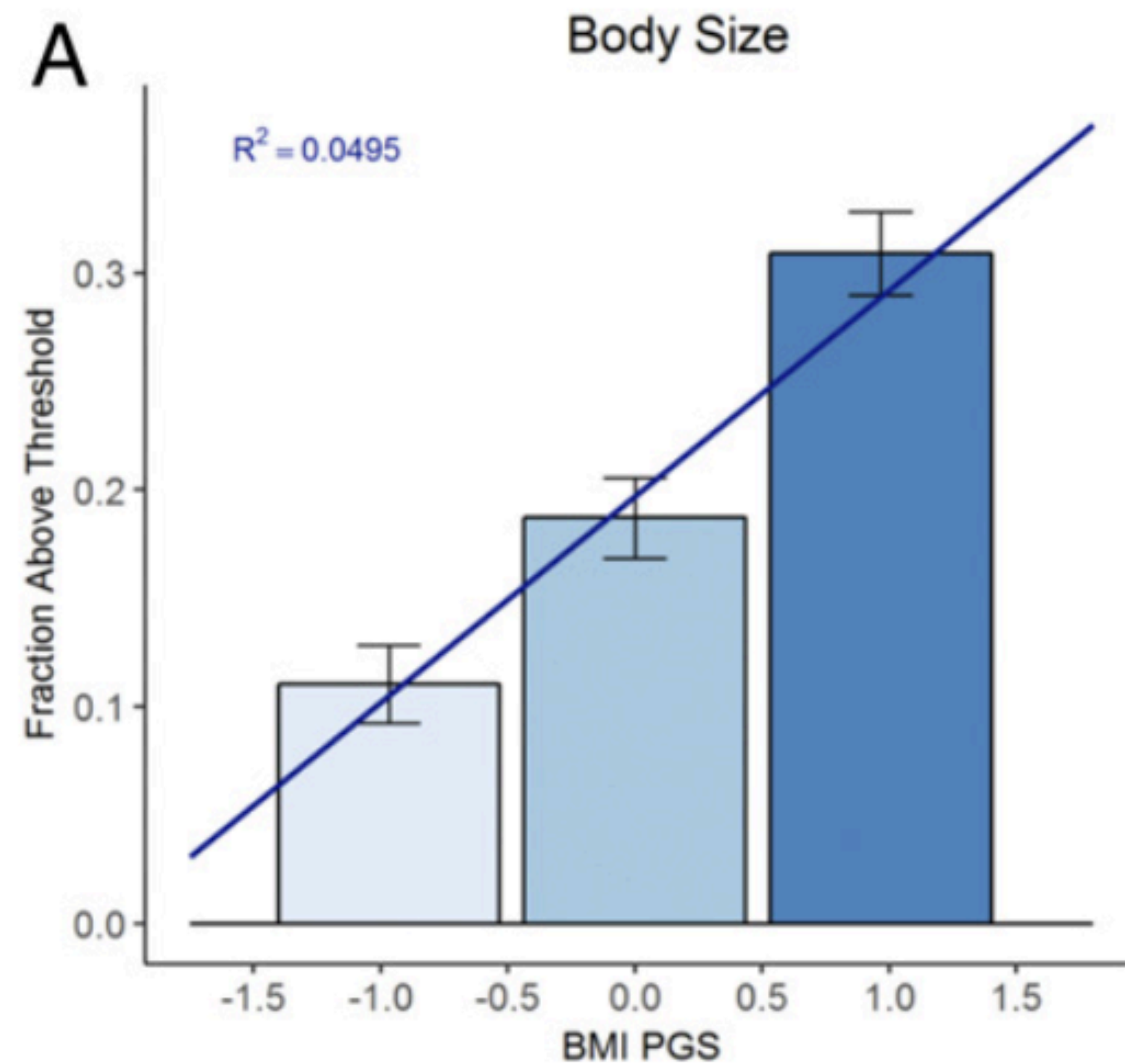
Education can reduce health differences related to genetic risk of obesity

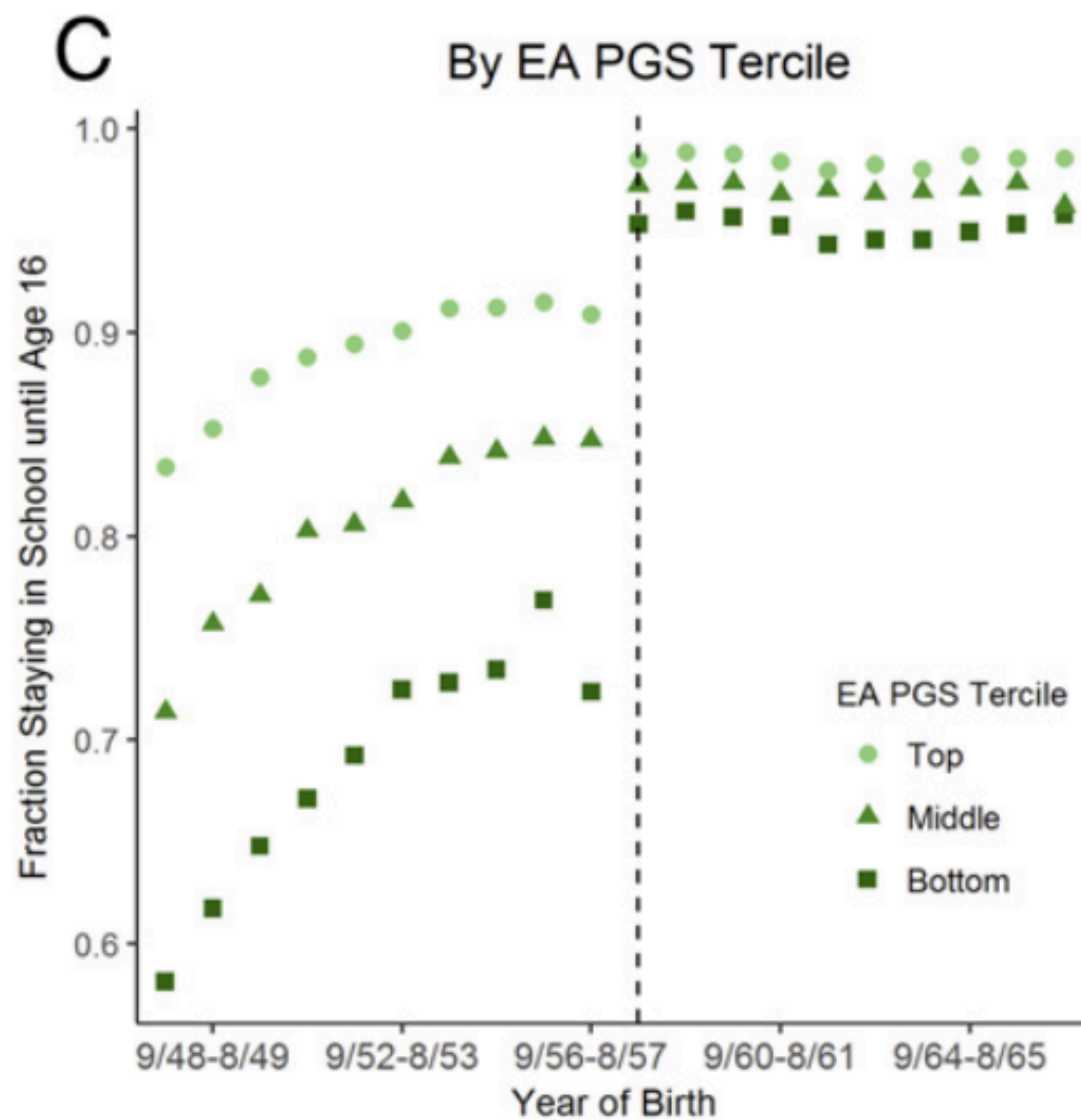
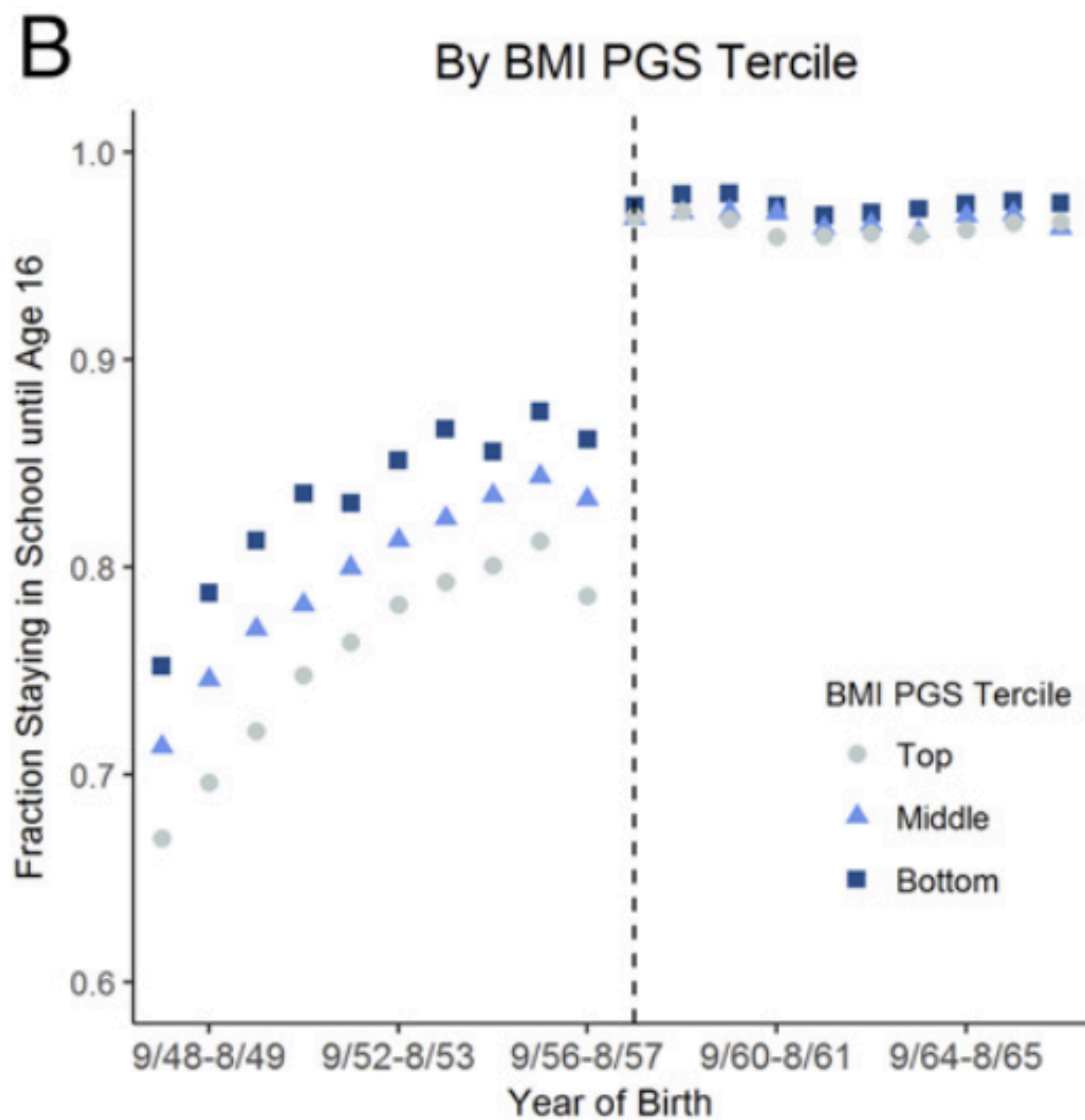
Silvia H. Barcellos^{a,1}, Leandro S. Carvalho^a, and Patrick Turley^{b,1}

^aCenter for Economic and Social Research, University of Southern California, Los Angeles, CA 90089; and ^bAnalytic and Translational Genetics Unit, Massachusetts General Hospital, Boston, MA 02114

Edited by Kenneth W. Wachter, University of California, Berkeley, CA, and approved September 4, 2018 (received for review February 16, 2018)

Genetic predisposition for obesity





$$\begin{aligned}
Health_i = & \beta_0 + \beta_1(Edu16_i \times PGS_i) + \beta_2 Edu16_i + \beta_3 PGS_i \\
& + f(DoB_i) + (Edu16_i \times PC_i')\beta_4 + PC_i'\beta_5 + x_i'\beta_6 + u_i,
\end{aligned}
\tag{11}$$

Table 1. Effect of staying in school until age 16 on health indices

	Above threshold				Continuous			
	Body size	Lung function	Blood pressure	Summary index	Body size	Lung function	Blood pressure	Summary index
Interaction with BMI PGS								
BMI PGS $\times$ Edu16	−0.057*** (0.011)	−0.037** (0.016)	0.016 (0.018)	−0.101*** (0.016)	0.028 (0.033)	−0.091** (0.044)	0.073** (0.036)	−0.010 (0.042)
Edu16	−0.060* (0.035)	−0.089* (0.048)	0.106** (0.051)	−0.058 (0.050)	−0.119 (0.096)	−0.147 (0.124)	0.118 (0.102)	−0.092 (0.122)
BMI PGS	0.124*** (0.010)	0.048*** (0.015)	0.025 (0.016)	0.152*** (0.015)	0.263*** (0.030)	0.127*** (0.040)	0.024 (0.033)	0.206*** (0.038)
<i>P</i> value for H_0 : no effect of education	7.97×10^{-10}	6.09×10^{-5}	0.004	1.50×10^{-14}	0.455	0.002	0.002	0.545

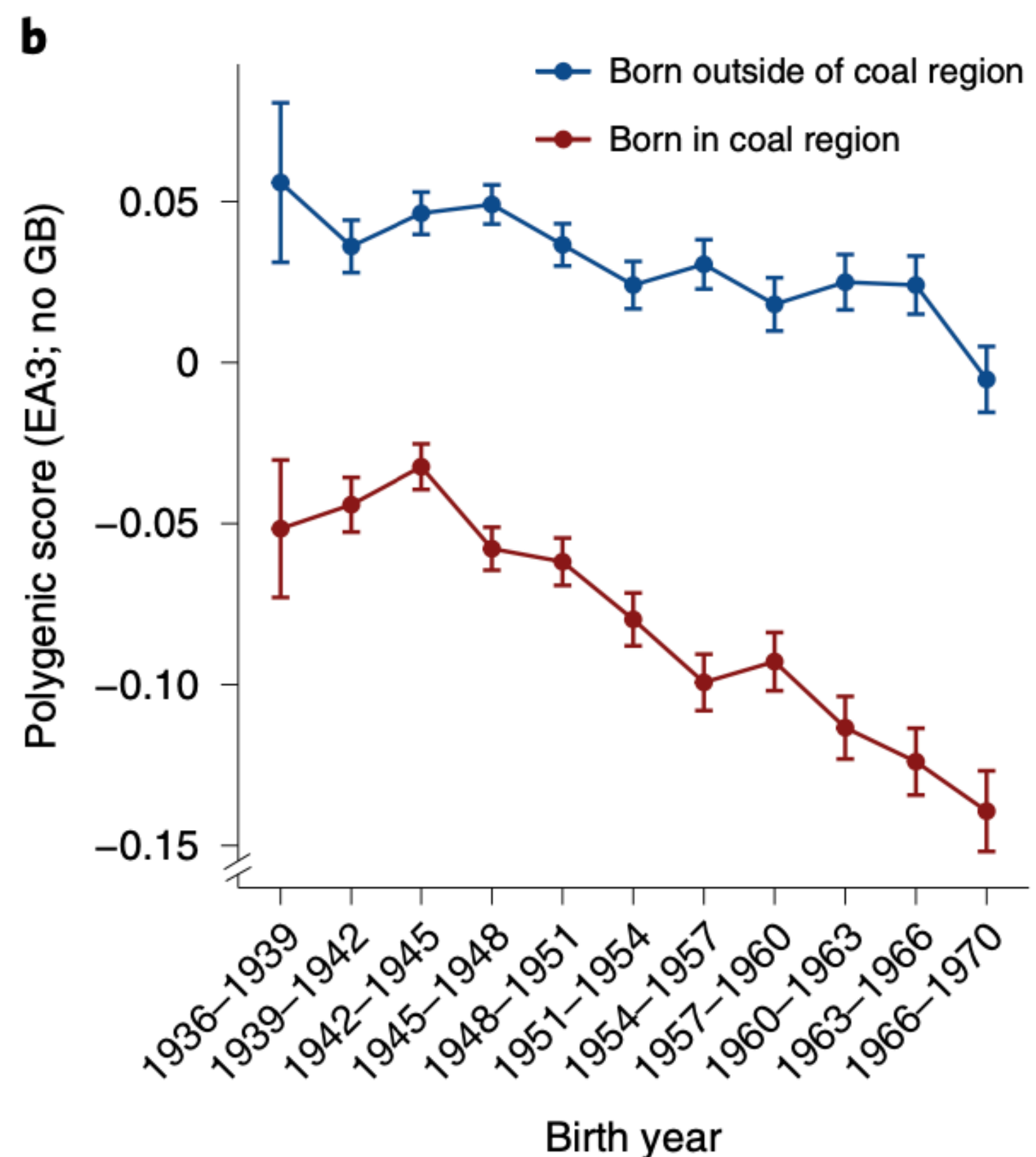
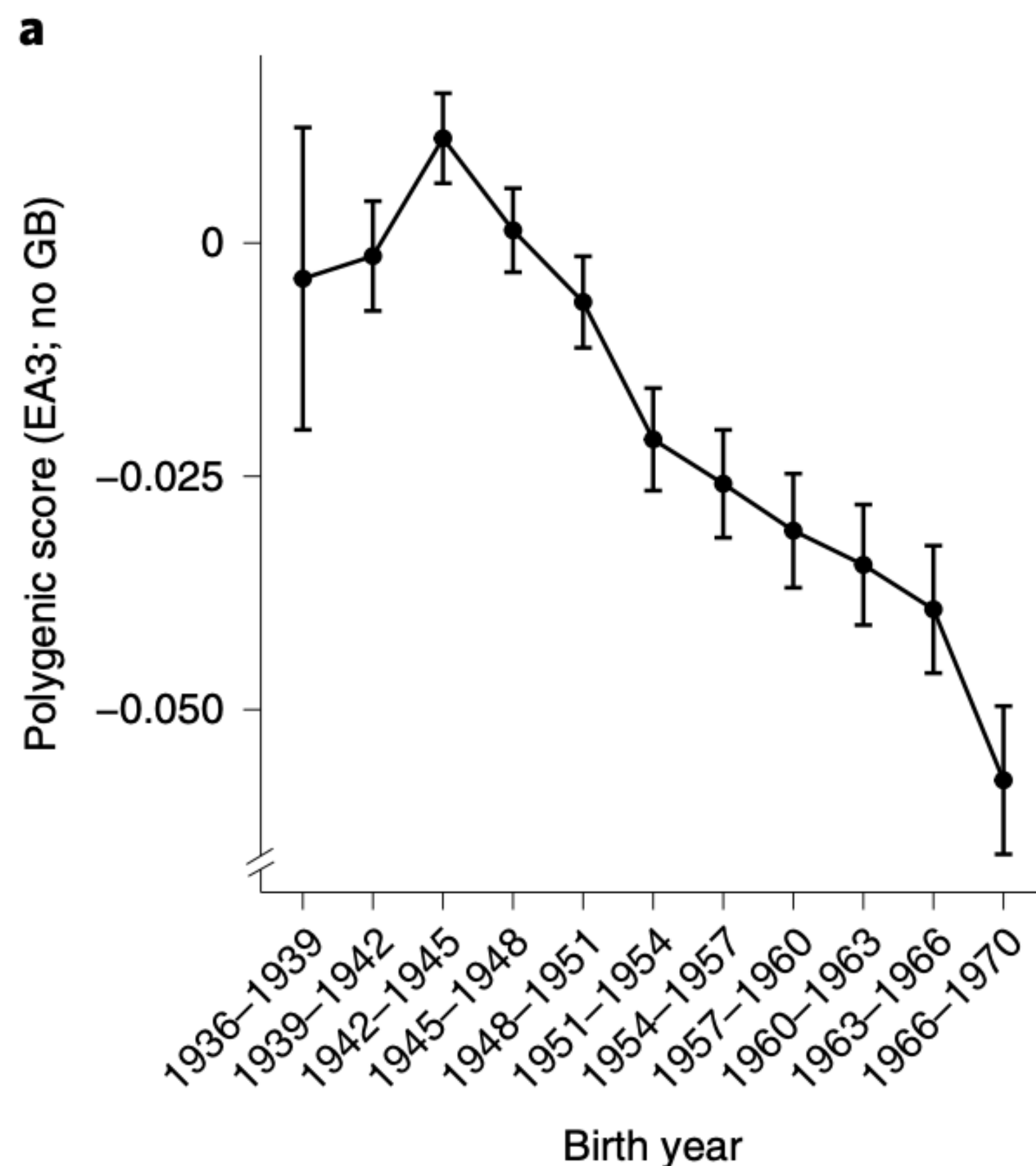
Significance

- *“Our findings show that the effects of education on health were not uniform across genetic backgrounds, benefitting those with greater genetic risk for obesity more. In other words, education not only affected health, corroborating the social determinants hypothesis, but it also reduced the role played by genetic factors: The association between genetic predisposition to obesity and unhealthy body size was reduced among cohorts who were forced to stay in school longer”*

Exogenous G? The role of rGE

- Assuming that G and E are independent and that G are exogenous might be problematic
- Evidence of correlation between PGS and possible environments

PGS education and birth year



Geographical distribution PGS

EA2 (no UK Biobank)



Genetic correlates of social stratification in Great Britain

Abdel Abdellaoui^{1*}, David Hugh-Jones², Loic Yengo³, Kathryn E. Kemper³, Michel G. Nivard⁴, Laura Veul¹, Yan Holtz³, Brendan P. Zietsch⁵, Timothy M. Frayling⁶, Naomi R. Wray^{3,7}, Jian Yang^{3,7}, Karin J. H. Verweij¹ and Peter M. Visscher^{3,7*}

Human DNA polymorphisms vary across geographic regions, with the most commonly observed variation reflecting distant ancestry differences. Here we investigate the geographic clustering of common genetic variants that influence complex traits in a sample of ~450,000 individuals from Great Britain. Of 33 traits analysed, 21 showed significant geographic clustering at the genetic level after controlling for ancestry, probably reflecting migration driven by socioeconomic status (SES). Alleles associated with educational attainment (EA) showed the most clustering, with EA-decreasing alleles clustering in lower SES areas such as coal mining areas. Individuals who leave coal mining areas carry more EA-increasing alleles on average than those in the rest of Great Britain. The level of geographic clustering is correlated with genetic associations between complex traits and regional measures of SES, health and cultural outcomes. Our results are consistent with the hypothesis that social stratification leaves visible marks in geographic arrangements of common allele frequencies and gene-environment correlations.

Distribution of G

- The distribution of our genes depends of the pattern of mating, admixture and migration of our ancestors.
- Also, PGS depends highly on GWAS effects that are estimated in specific population
- Also, Sometimes cohorts used for GWAS are composed by people in old age who survived a certain age and agree to be genotyped and fill a questionnaire, hence it could be selected.

Why not a Gwas with GXE?

- Number of parameters to estimate would be too big to analyse with the current techniques

Status of GxE research

- Most of the “sociogenomics research” is about GxE
- Policy implications! We cannot change G, but we can do a lot for E
- Implications for inequality
-

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Three phases of **Gene× Environment interaction** research: Theoretical assumptions underlying **gene** selection

X Zhang, J Belsky - Development and Psychopathology, 2022 - cambridge.org

Some **Gene × Environment interaction** (G×E) research has focused upon single candidate genes, whereas other related work has targeted multiple genes (eg, polygenic scores). Each ...

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[HTML] **Gene-environment interaction** analysis of redox-related metals and genetic variants with plasma metabolic patterns in a general population from Spain: The ...

M Galvez-Fernandez, F Sanchez-Saez... - Redox Biology, 2022 - Elsevier

Background Limited studies have evaluated the joint influence of redox-related metals and genetic variants on metabolic pathways. We analyzed the association of 11 metals with ...

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Pathogenesis of chronic obstructive pulmonary disease: understanding the contributions of **gene–environment** interactions across the lifespan

A Agustí, E Melén, DL DeMeo... - The Lancet Respiratory ..., 2022 - Elsevier

... **gene–environment interaction** occurs and the cumulative history of previous **gene–environment** interactions between genes (G) and the **environment** (E) by integrating information from ...

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Vulnerability to early stressful life events: The moderating role of HOMER1 g in the **gene× environment interaction**

Y Chen, L Wang, D Jiang, S Zhu... - European Journal of ..., 2022 - Taylor & Francis

... Therefore, candidate **gene × environment** (cG × E) **interaction** analysis considering environme

... traits in **gene–environment** interactions, but more research on this **interaction** is being ...

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